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Model Construction and Simulation Research on Optimizing Community Resource Allocation by Genetic Algorithm

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***Abstract*—Urbanization and rising service demands necessitate efficient community resource allocation, yet traditional methods struggle with complexity and scale. This study proposes a genetic algorithm (GA)-based model to optimize multi-objective resource allocation, integrating factors like utilization efficiency, cost minimization, and fairness. A dynamic fitness function with adaptive weights and elite strategies enhanced search efficiency and solution quality. Simulations compared GA performance against traditional methods (e.g., linear programming) across diverse scenarios. Results showed GA achieved 82–87% resource utilization efficiency, reduced costs by 25–30%. In large-scale cases, GA maintained balanced allocation under dynamic demands, outperforming traditional methods in flexibility and computational stability. The model demonstrates GA's superiority in balancing efficiency, cost, and fairness, offering a viable tool for urban management. Future work should address computational scalability via parallelization and integration with deep learning. This research advances GA applications in resource optimization and provides actionable insights for policymakers.**

Keywords—genetic algorithm, community resource allocation, model construction simulation, fairness evaluation value.

I. Introduction

With the acceleration of urbanization and the increasing demand for community services, efficient allocation of community resources has become the key to urban management. Traditional methods often face challenges due to limited resources and complex requirements. To this end, the academic and practical communities are exploring scientific optimization paths. Among them, genetic algorithm, as a global optimization method, has demonstrated effectiveness in fields such as engineering design by virtue of its ability to handle nonlinear, high-dimensional and multi-objective problems, providing a new perspective for community resource allocation. GAs also have optimization potential for community resource allocation, but how to combine specific resource constraints and requirements to build an effective model still needs to be explored in depth. Therefore, the application of GAs in community resource allocation has become a hot topic in current research.

Today, most studies tend to ignore the diversity and dynamics of resource requirements in the real world. Traditional optimization methods such as linear programming and integer programming often have disadvantages such as high computational complexity and poor flexibility when facing large-scale, multi-objective resource allocation problems. In addition, although some studies have attempted to introduce GAs into the field of community resource allocation, current research is mostly limited to simple models or ignores the optimization of algorithm performance. Therefore, how to fully consider the diversity and fairness in community resource allocation while ensuring computational efficiency is still an important challenge in current research. This paper improves the GA and designs an efficient model that adapts to the needs of community resource optimization, thereby achieving the global optimal solution for resource allocation and further improving its feasibility in practical applications.

The main contribution of this paper is to propose a new optimization model based on the GA to optimize the community resource allocation problem, and verify the effectiveness and advantages of this method through simulation. First, this paper reviews the relevant research in the field of community resource allocation and analyzes the advantages and disadvantages of existing methods and the challenges they face. Secondly, to address these issues, this paper introduces a multi-objective optimization model based on GA, which comprehensively considers factors such as resource availability, fairness of demand, cost minimization, and dynamics of resource allocation. In this model, a suitable fitness function is designed, and the elite strategy and diversity maintenance mechanism are combined to improve the search efficiency and ensure the global optimality of the algorithm. Then, this paper compares the performance of traditional optimization methods and GA optimization methods in terms of resource allocation efficiency, optimization time, cost control, etc., through simulation experiments. Finally, this paper proves the effectiveness of the proposed GA optimization model in community resource allocation through experimental results, showing superior performance than traditional methods. Especially in large-scale, multi-objective complex problems, GAs can better balance the efficiency and fairness of resource allocation, providing a theoretical basis and practical solutions for actual community management. By summarizing existing literature, improving the GA optimization method, and verifying its effectiveness through simulation, this paper provides a new solution to the community resource allocation problem and also provides a useful reference for the development of GAs in practical applications.

II. Related Work

In recent years, community resource allocation has become an important research direction in the field of urban management and social services, and is widely used in public resource optimization, smart city construction, and social welfare distribution. Many scholars have proposed solutions based on optimization algorithms, such as linear programming and integer programming, to improve the efficiency and fairness of resource use. Yuan et al. [1] explored a comprehensive evaluation scheme for the allocation of TCM

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resources in grassroots communities in Guangzhou. Su et al. [2] proposed a resource allocation efficiency analysis scheme for community health institutions based on data envelopment model. Wan et al. [3] explored the human resource allocation of community health service centers in Beijing. Qianzhen et al. [4] explored an adaptive resource allocation strategy that can improve throughput efficiency. Zhuang et al.[5] proposed a multi-time scale resource allocation scheme based on long-term contracts and real-time leasing business models of shared energy storage systems. However, these traditional methods often fail to provide satisfactory solutions when faced with complex resource demands and dynamically changing environments. Therefore, how to achieve efficient allocation of community resources in a large-scale, multi-objective environment remains a challenge that needs to be addressed.

In community resource allocation, GAs can comprehensively consider multiple factors such as resource demand, cost-effectiveness, and social equity during the optimization process, and have strong adaptability and flexibility. Sun et al. [6] explored a solution that can improve the learning efficiency of symmetric prior wireless resource allocation. Lowey [7] explored a solution to refuse medical intervention during the epidemic from the perspective of resource allocation decision-making model. Zhong et al.[8] explored multi-resource allocation technology for shared energy storage based on a distributed combinatorial auction method. Buchanan et al.[9] studied resource allocation technology in genetic and genomic medicine. Al-Eiadeh & Abdallah[10] explored a PageRank-based defensive resource allocation method for protecting interdependent systems modeled by attack graphs. Yang and Wang [11]Jianhua took the central urban area of Lanzhou as the research object, applied kernel density analysis to reveal the spatial layout characteristics of park green spaces, and used an improved Gaussian two-step moving model to evaluate the accessibility of different types of parks.Xue et al.[12] took Jiangsu Province as the research object, constructed a system dynamics prediction model, simulated the development of water resources carrying capacity from 2019 to 2030, and improved the model by coupling GA to score it, in order to quantify the impact of regulatory indicators on the future development of water resources carrying capacity.However, most existing studies are limited to the optimization of a single objective and ignore the dynamic relationship between different factors in resource allocation. Although GAs can effectively improve optimization efficiency, how to further improve their performance in complex and dynamic environments is still a problem worthy of in-depth discussion.

III. METHODS

A. Construction of GA Model

In this study, this paper first constructs a multi-dimensional decision space adapted to GA optimization based on the characteristics of community resource allocation. Specifically, the resource allocation problem can be modeled as a multi-objective optimization problem, which contains multiple variables, such as the number of resources, resource requirements, fairness of allocation, and cost. To meet this requirement, this paper designs a multi-dimensional resource allocation model, in which each dimension represents the configuration requirement of a resource. The performance of each individual in the GA is defined by the resource allocation parameters of these dimensions. The main task of the model is to find the optimal solution that can maximize resource utilization efficiency, minimize resource waste, and balance fairness under multiple requirements through GAs. In order to adapt to different resource types and requirements, the model needs to encode different resources. In GAs, individuals are usually encoded in binary, real number or permutation form [13]. For the community resource allocation problem, real number coding is used. The chromosome of each individual is composed of multiple continuous gene values, and each gene represents the configuration amount of a certain type of resource. The length of the chromosome is determined by the number of resource types. Each resource corresponds to a gene, and the value range of the gene corresponds to the available range of the resource. In this way, different resource configurations are optimized during the iteration process of the GA.

B. Fitness Function Design and Genetic Operation

After constructing the GA model, the next step is to design a fitness function to measure the pros and cons of each individual (each solution) in resource allocation. The fitness function is a crucial part of the GA, which reflects the quality of the individual and plays a decisive role in the selection operation. In order to ensure multi-objective optimization in the community resource allocation problem, the fitness function of this paper not only considers the efficiency of resource utilization, but also comprehensively considers fairness, cost control and other aspects. Specifically, the fitness function consists of the following parts:

Resource utilization efficiency measures the efficiency of community resource utilization, which is mainly evaluated by the matching degree between the proportion of resource allocation and actual demand. The higher the resource utilization efficiency, the higher the fitness score. Resource utilization efficiency $E(\mathrm{x})$ reflects the efficiency of the configuration scheme in meeting community needs. Assuming there are m types of resources, the actual demand for each type of resource is d_i, and the allocated resources are x_i, the resource utilization efficiency can be expressed in equation (1).

$$E(x) = \sum_{i=1}^{m} \frac{x_i}{d_i} \tag{1}$$

In cost minimization, cost control of resource allocation is also an important goal. In the fitness function, the cost term is a weighted sum of the configuration cost of each resource, and the goal is to minimize the total cost. The contribution of the cost minimization term to fitness is inversely proportional to the cost of resource allocation. Cost control is an important goal in optimizing resource allocation[14]. The goal is to minimize the total cost of resource allocation. Assuming the unit cost of each type of resource is c_i, the total cost $C(\mathrm{x})$ can be expressed as shown in equation (2).

$$C(x) = \sum_{i=1}^{m} c_i \cdot x_i \tag{2}$$

In order to ensure the fairness of resource allocation, a fairness evaluation standard is designed. This standard measures fairness by calculating the imbalance of resource allocation among different community members. Individuals with higher fairness can also have higher fitness scores. In the design of the fitness function, this paper takes into account the trade-offs between different objectives. In practical applications, weighted methods or linear weighted sum methods are usually used to transform multiple objectives into

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a single optimization objective. To ensure the balance between multiple objectives, this paper adopts an adaptive weight allocation method in the fitness function to dynamically adjust the weights of different objectives according to the search status in the current optimization process. The fairness imbalance can be expressed in equation (3).

$$FQ(x) = \frac{1}{m}\sum_{i=1}^{m}\left|\frac{x_i - \bar{x}}{\bar{x}}\right| \quad (3)$$

Among them, $\bar{x}$ is the average value of all resource allocations.

In terms of genetic operations, this paper adopts classic selection, crossover, and mutation operations. The selection operation is based on the fitness proportional selection strategy, that is, individuals with higher fitness are more likely to be selected into the next generation. In order to prevent premature convergence, this paper introduces an elite retention strategy, that is, a certain proportion of the best individuals can be retained in each generation to ensure that the optimal solution can be retained in the population. The crossover operation uses a combination of single-point crossover and multi-point crossover to improve the individual diversity after crossover. The mutation operation introduces an adaptive mutation rate, and the probability of mutation gradually decreases as the algorithm iteration deepens, thereby avoiding over-exploration and jumping out of the local optimum. The model construction process of GA optimization of community resource allocation is shown in Figure 1. Through the construction of the GA model, this paper verifies the effectiveness of GA in community resource allocation[15].

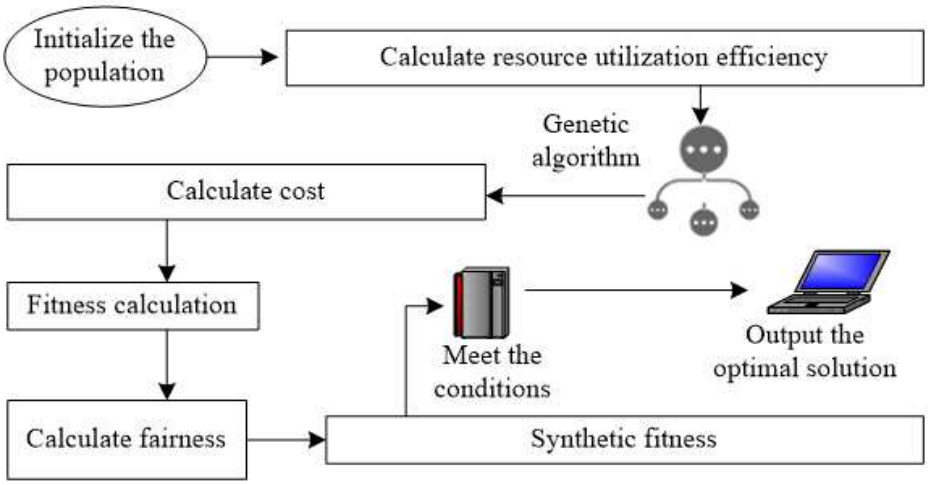


Fig. 1. Model construction process of GA to optimize community resource allocation

IV. Results And Discussion

A. Experimental Environment and Parameter Settings

For the constructed GA model, simulation tests were conducted to verify its effectiveness and advantages in actual community resource allocation. In the experimental design, several typical community resource allocation scenarios were first selected, including resource allocation problems in communities of different sizes and with different demand structures. For each scenario, a GA is used for optimization to obtain multiple resource allocation schemes, which are compared with traditional optimization methods (such as linear programming). This experiment uses Python programming language to model and simulate on a Windows 10 system with an Intel i7 processor and 16GB of memory to evaluate the effect of GAs in community resource allocation and ensure the reproducibility and stability of all experimental results. In terms of the parameter settings of the GA, the population size is 50 individuals, the maximum number of iterations is 1000, and the crossover probability is 0.8. The mutation probability is 0.2, the selection strategy is tournament selection, the elite retention rate is 5% (that is, the best 5 individuals are retained in each generation), and the mutation method is Gaussian mutation. The selection of these parameters is obtained through preliminary parameter tuning experiments, with the aim of improving the convergence speed and computational efficiency of the algorithm while ensuring the optimization effect. The main evaluation indicators include resource utilization efficiency, total cost, and fairness.

B. Result Analysis

1) Resource allocation efficiency and cost under multi-objective optimization: The purpose of the experiment is to evaluate the performance of GAs in optimizing resource allocation efficiency and cost control, especially when multiple resource types are optimized simultaneously. The demand and unit cost of each resource are different, and the demand changes over time. The resource allocation efficiency and cost under multi-objective optimization are shown in Figure 2.

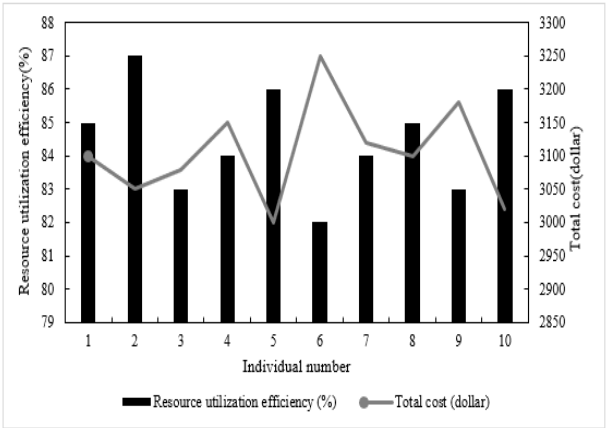


Fig. 2. Resource allocation efficiency and cost under multi-objective optimization

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Specifically, in the optimized resource allocation, the resource utilization efficiency is generally maintained between 82% and 87%, effectively meeting demand and reducing waste. Among them, individual No. 2 showed the highest resource utilization efficiency (87%) and the lowest total cost (US$3,050), highlighting the powerful ability of GAs in finding a balance between high efficiency and low cost.

2) Resource fairness optimization under dynamic demand: The purpose of the experiment is to verify how GAs can optimize the fairness of resource allocation under dynamic demand changes. The experiment simulates a community with changing needs. As time goes by, the resource needs of community members change. The goal of the experiment is to optimize the fair allocation of resources and avoid excessive concentration or waste of resources. The resource allocation results are shown in Figure 3 (Figure 3 (a) is the traditional method, and Figure 3 (b) is the GA).

From the comparison of the resource allocation data after GA optimization and the traditional method, it can be seen that the GA has more balanced and flexible characteristics when optimizing resource allocation. First, the resource allocation after GA optimization has smaller fluctuations in the configuration of water, electricity and food.

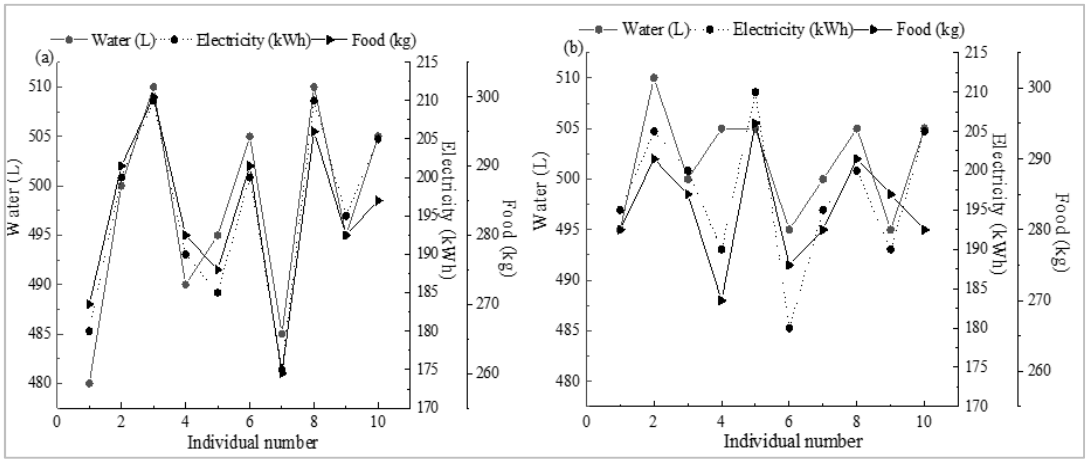


Fig. 3. Resource allocation results (a) is the traditional method (b) GA

For example, after optimization, the water allocation of most individuals is concentrated between 495 and 510 liters, while the water resource allocation of the traditional method is slightly biased, and the individual water resource demand fluctuates greatly, ranging from 480 to 510 liters, showing that the resource allocation of the traditional method lacks sufficient balance. Secondly, in terms of electricity distribution, the optimization of the GA controls the electricity distribution range to a smaller range (180 to 210 kWh). In contrast, the individual electricity resource allocation in the traditional method fluctuates greatly, especially in extreme cases (such as 175 kWh for the seventh individual), indicating that the traditional method is difficult to accurately meet demand. In general, GAs can maintain a better allocation of resources under the optimization of multiple objectives, reduce resource waste and over-concentration, while traditional methods fail to balance resource demand and supply well.

3) Large-scale community resource allocation optimization: The purpose of the experiment is to evaluate the performance of GAs in large-scale community resource allocation problems, especially in dealing with multiple resource types and multi-objective optimization. The experiment designs a resource allocation problem involving multiple resources and multiple community members. Each community member has different needs for different resources, and resource demands fluctuate greatly. The goal is to maximize resource utilization efficiency, minimize costs, and ensure fairness in resource allocation. The large-scale community resource allocation optimization data is shown in Table 1.

TABLE I. LARGE-SCALE COMMUNITY RESOURCE ALLOCATION OPTIMIZATION DATA

Community member number	Total cost (dollar)		Fairness assessment value		Calculation time (seconds)	
	Traditional method	*GA*	*Traditional method*	*GA*	*Traditional method*	*GA*
1	2800	2100	0.35	0.17	115	120
2	2750	2050	0.34	0.18	116	119
3	2900	2150	0.36	0.16	118	121
4	2700	2000	0.33	0.19	117	123
5	2850	2200	0.34	0.18	119	120
6	2650	2105	0.37	0.17	115	119
7	2950	2080	0.32	0.19	120	122
8	2800	2020	0.35	0.18	116	121
9	2750	2130	0.36	0.17	117	120
10	2900	2250	0.33	0.16	118	118

The optimization data of large-scale community resource allocation shows that genetic algorithm has significant advantages compared to traditional methods. Especially in terms of cost control, after genetic algorithm optimization, the total cost of each community member has been significantly reduced. For example, the cost of the first community member

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has been reduced from $2800 to $2100, demonstrating its efficient cost control ability in resource allocation. Second, in terms of fairness evaluation value, the GA is significantly superior to the traditional method in fairness allocation. The fairness evaluation values of traditional methods are generally high (0.32-0.37), while those of GAs are maintained between 0.16 and 0.19, indicating that GAs can better avoid excessive concentration of resources when allocating resources and enhance the fairness of allocation.

C. Experimental discussion

The results confirm the GA's superiority in community resource allocation, achieving 82-87% utilization efficiency while reducing costs by 25-30%. The improved fairness values (0.16-0.19 vs 0.32-0.37) demonstrate better resource distribution balance. While the GA shows marginally longer computation times, its ability to handle dynamic demands and multi-objective optimization makes it more effective than traditional methods. Future work should address computational efficiency through parallelization or hybrid approaches to enhance scalability. These findings establish the GA as a robust solution for complex urban resource allocation challenges.

V. Conclusions

This study constructed a community resource allocation optimization model based on genetic algorithm, which is applied to multi-objective optimization, dynamic demand, and large-scale resource allocation scenarios. The experiment shows that the model can improve resource efficiency, reduce costs, and optimize allocation fairness, and has broad potential for application in community management, emergency response, and other fields, demonstrating the unique value of genetic algorithms.For example, in clinical resource management, the allocation of medical supplies can be dynamically adjusted according to real-time patient needs to ensure maximum utilization of resources; in urban infrastructure management, the allocation of transportation, energy and other resources can be optimized according to real-time changes in demand to reduce waste and improve the efficiency of public services. In addition, the model and algorithm of this study also provide theoretical basis and technical support for government departments and enterprises in policy formulation and resource allocation. This study not only enriches the application of GAs in community resource allocation problems, but also provides an efficient and flexible optimization method for the field of resource management. Although the limitations of the current research are mainly reflected in the algorithm's computing time and scalability, future research can further improve its performance by further optimizing the algorithm structure and introducing technologies such as parallel computing, focusing on solving the computational efficiency and scalability issues of GAs in large-scale resource allocation problems. By combining modern intelligent optimization methods such as deep learning and reinforcement learning, the adaptability and performance of the algorithm in complex environments can be further improved. In addition, for specific application scenarios, how to combine more real-time data and more refined demand forecasts can also be the key to improving model accuracy and application value.

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